

The Rise of Dollar Credit in Emerging Market Economies and US Monetary Policy

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Abstract

This paper uses a multivariate correlated unobserved component model to study the dynamic relationship among dollar credit in emerging market economies, US interest rates and the dollar index. This approach decomposes each series into permanent and transitory components and also allows us to estimate the correlation between the permanent movements and transitory movements of each series. The results from this model suggest that the transitory shocks to dollar credit in emerging market economies are highly negatively correlated with the transitory shocks to the US interest rate and the transitory shocks to the dollar index. The estimate of the cyclical component of the dollar credit in emerging market from our model captures the recent boom and bust phase in this market and compares favorably to a univariate trend-cycle decomposition benchmark.

Keywords: Dollar Credit, Emerging Market Economies, Monetary Policy Spillover, Unobserved Component Model, Trend and Cycle Decomposition, Kalman Filter

JEL classification: E32, E51, E58, F32, F34, F36, G15

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1 Introduction

The surge of dollar credit in emerging market economies (EMEs henceforth) has drawn close attention from the international financial market participants and policy makers due to its potential role in monetary and financial stability of the global economy. The data from the Bank of International Settlements shows that the outstanding US dollar credit to non-bank borrowers in EMEs has risen from 1.69 trillion dollar in 2008Q4 to 3.25 trillion dollar in 2015Q4. The cyclical nature of EME dollar credit¹ could also amplify the overall credit conditions in the domestic economy and increase the vulnerability of these EMEs in dealing with negative economic shocks. For example, during the expansionary phase, the dollar credit floods into EMEs, which raises the challenge to stabilize the domestic monetary base. In the contractionary phase, the dramatic outflow of dollar credit also creates difficulty to stabilize the exchange rate and asset prices (Avdjiev et al., 2012). This was very evident during the 'taper tantrum' of 2013. On the other hand, it has also been argued that the increasing role of dollar credit in EMEs may signify a more integrated global financial market and a greater degree of risk sharing across different countries.

The growing academic interest in the literature on examining the dollar credit in EMEs has also coincided with the recent emphasis on the role of US monetary policy in boosting global liquidity. Global liquidity refers to the global factor that drives cross-border spillover in financial conditions and credit growth. Shin (2013) proposes two phases of global liquidity after the Millennium. The first phase of global liquidity transmission (2002-2008) is associated in form of bank loans through global banking system. The international bond market gradually took over the share of bank loans in the second phase of global liquidity transmission starting in 2009. Since then, the large scale of bond purchases as a result of Quantitative Easing (QE) programs had led to the portfolio rebalancing effect in the international bond market. A few papers and policy studies have provided narrative evidence (Borio, et al. 2011; McCauley, et al., 2015) in support of the hypothesis that the abundance of global liquidity was one of the causes of the boom in the dollar credit in the EMEs.

¹EME dollar credit in this paper refers to the outstanding dollar credit to non-bank borrowers in emerging market economies.

Although much attention has been given to the rise of dollar credit and its apparent relationship with the US monetary policy and the valuation of the US dollar, not much work has been done to systematically untangle the short-run and the long-run relationships among these variables. It is perfectly plausible to think that the short-run relationship between dollar credit in emerging markets and the stance of monetary policy in the US may be very different than its long-run relationship. The long-run dollar credit in the emerging market may be affected more by its long-run absorptive capacity instead of the short-run increase in liquidity or the weakness of the US dollar. Therefore, it is very important to decompose the overall EME dollar credit and isolate the cyclical variations from its long-term trend, by taking into account the information on US interest rate and the dollar valuation. In addition to estimating the short-run and the long-run correlation between the shocks to each series, this approach will also yield us a quantitative estimate of how big the cyclical component in dollar credit was at different points in time during the last few years.

To examine the long-run and short-run relationship among these variables, we propose to use a correlated multivariate unobserved component (UC) model. This model allows us to decompose the movements in dollar credit, interest rate and dollar index² into a slow-moving trend component and a cyclical component. The slow-moving trend captures the long-run evolution of these variables and the cyclical component captures the short-run movements. This model allows us to not only estimate the permanent and transitory movements, but also provides us a measure of correlation between the long-run and the short-run movements of dollar credit, interest rate and dollar index. For example, if the recent narrative about the role of very low interest rate in EME dollar credit is correct, then we would observe a negative correlation between the shock to the transitory component of dollar credit and the shock to transitory component of the US interest rate.

We find interesting and economically meaningful results from the estimated multivariate correlated unobserved component model. The maximum likelihood estimates of correlated multivariate UC model suggest that there is a strong negative correlation between the transitory shock to dollar credit and the transitory shock to the US interest rates. We also find a very high negative correlation

²Dollar index refers to the broad trade-weighted US dollar index. This variable is a proxy of the dollar valuation against other currencies.

between the shocks to dollar credit and the shocks to the dollar index, implying an appreciation of US dollar is associated with a decline in dollar credit in the EMEs in the short-run. These results support the anecdotal and narrative evidence on the strong relationship between dollar credit and the US interest rates and also between the dollar credit and the strength of the dollar. We also find that the trend-cycle decomposition of the EME dollar credit from our multivariate correlated unobserved component model captures the recent boom and bust behavior and compares favorably to a univariate trend-cycle decomposition benchmark. In particular, our results suggest that the dollar credit before the taper tantrum was 10% above its long-run trend in the emerging market economies.

The rest of the paper are structured as follows. The second section provides a literature review about global liquidity transmission and discusses the associated monetary policy spillover effect. The third section introduces the data used in this study and the setup of the correlated multivariate UC model. The fourth section interprets and presents the results from the model. The last section concludes the paper.

2 Literature Review

The literature on dollar credit in EMEs is nascent. Few paper have tried to address this issue from different perspectives. In understanding the phenomenon that dollar credit outside the US behaves differently from the US domestic dollar credit, Borio, McCauley and McGuire (2011) take a look at the recent behavior of international credit and associate it with the overall credit condition. Their descriptive analysis reveals the fact that US dollar credit in some countries has been outgrowing the overall credit during the credit booms. A formal analysis later on from Avdjiev, McCauley and McGuire (2012) regresses cross-sectional change in credit-to-GDP ratio on change in international credit during the credit boom (2002-2008) and regresses credit growth in EMEs on change in international credit during the credit bust (2008-2011). Their results suggest that international credit amplifies the overall credit cycles in EMEs. While in these studies, the authors often use the share

of international credit in the overall credit to measure the cyclical variation of international credit, it is worth to recognize that the trend of international credit and the trend of the overall credit can be driven by different underlying factors, therefore do not need to share the common trend in the numerator and denominator. For instance, the long-term trend of international credit could be explained by the integration of international financial market, but the long-term trend of the overall credit condition may respond more to the domestic economic fundamentals. Thus without isolating the long-term trend from these two credit series, it is very difficult to perform a clean analysis about the cyclical co-movement between international credit and the overall credit condition.

Since the recent global financial crisis of 2008-2009, the outstanding US dollar credit to non-bank borrowers in EMEs has roughly doubled within the past seven years. The commentators have also associated this surge to US monetary policy spillover, mainly because of ultra low interest rates in the US. He and McCauley (2013) survey a number of studies on the transmission of monetary policy of the major advanced economies to East Asia and conclude that policy rates, bond yields and currency appreciation are three price channels in the transmission process. McCauley, McGuire and Suchko (2015) also link the US monetary policy, leverage and flow into bond funds to explain dollar credit extended to non-US borrowers. In this paper, since we the goal is to understand the behavior of the overall dollar credit condition, regardless of the credit instruments, we focus on the two price channels—interest rate and currency appreciation/depreciation, an analytical framework which can be interpreted as based upon international interest rate parity theory.

It is worth to point out that we focus on *EME dollar credit*, a subject we believe to be ideal to study the monetary policy spillover effect in global liquidity transmission. This is because, the composition of bank loan and bond issuance depends on country-specific contexts, for instance, the regulatory emphasis of capital control. Tighter regulation on international banking practice could push the domestic borrowers to issue bonds overseas and vice versa (Caballero et al., 2015). Since our goal is to understand the trend and cycle of dollar credit in emerging market economies, we purposely ignore the credit breakdown based on types of financial instruments. Furthermore, we choose to study EME dollar credit, instead of euro, yen or other currency credit, is because different

currency credit could be sensitive to different monetary policy rates and dollar credit dominates other currencies in the currency breakdown of international credit (Borio, et al., 2011). The last but not the least, we work with the dollar credit in EMEs, because dollar credit outside US may respond to factors in different ways when it comes to EMEs compared to advanced economies. (Borio, et al. 2011; McCauley, et al., 2015)

3 Data and Empirical Model

3.1 Data Description

Our study examines the dynamics of 3-variables: dollar credit in EMEs, US interest rate and dollar index. The period we study spans from the first quarter of 2000 to the last quarter of 2015, based on data availability.

The first variable is to measure dollar credit in EMEs. Bank for International Settlements (BIS) provides global liquidity indicators on the website to measure the ease of financing in global financial markets. Among various global credit aggregates, we use the US dollar credit to non-bank sector in EMEs in this study. The original data is measured in trillions of US dollar and we take the natural log of this series and use the log-transformed series in the model estimation.

The second variable is to measure the monetary policy rate in US economy. There are a few potential measures to choose from, for instance, three-month, one-year and ten-year treasury constant maturity rates. What one is more appropriate? The answer to this question depends on the term structure in credit instruments. EME dollar credit can be sensitive to short-term, medium-term and long-term interest rates since the series aggregate up all the credit instruments. Therefore, we propose to use the one-year constant maturity rate (in percent) to represent the US interest rate environment.

The third variable is to measure US dollar exchange rate in the global currency market. We use the broad trade-weighted US dollar index to measure the dollar valuation in the currency market. The larger the index, the stronger the US dollar is, or equivalently, the more US dollar appreciates,

and vice versa. We also log transform this variable in the model estimation to be consistent with the first variable measure.

The full data set contains 16-year quarterly observations, which means the sample size is 64 quarters. We use one-year burn-in period in this study, so the resulting trend and cycle series start from 2001Q1 to 2015Q4.

3.2 Empirical Model

In this paper, we use a tri-variate unobserved component model to decompose the trend and cycle dynamics in EME dollar credit (Y_t), monetary policy rate (U_t) and dollar index (A_t). In order to do so, we assume the following structure of the model.

EME Dollar Credit:

$$Y_t = \tau_{yt} + c_{yt} \quad (1)$$

$$\tau_{yt} = \mu_y + \tau_{yt-1} + \eta_{yt}, \eta_{yt} \sim N(0, \sigma_{\eta_y}^2) \quad (2)$$

$$c_{yt} = \phi_{1y}c_{yt-1} + \phi_{2y}c_{yt-2} + \varepsilon_{yt}, \varepsilon_{yt} \sim N(0, \sigma_{\varepsilon_y}^2) \quad (3)$$

Monetary Policy Rate:

$$U_t = \tau_{ut} + c_{ut} \quad (4)$$

$$\tau_{ut} = \mu_u + \tau_{ut-1} + \eta_{ut}, \eta_{ut} \sim N(0, \sigma_{\eta_u}^2) \quad (5)$$

$$c_{ut} = \phi_{1u}c_{ut-1} + \phi_{2u}c_{ut-2} + \varepsilon_{ut}, \varepsilon_{ut} \sim N(0, \sigma_{\varepsilon_u}^2) \quad (6)$$

Dollar Index:

$$A_t = \tau_{at} + c_{at} \quad (7)$$

$$\tau_{at} = \mu_a + \tau_{at-1} + \eta_{at}, \eta_{at} \sim N(0, \sigma_{\eta_a}^2) \quad (8)$$

$$c_{at} = \phi_{1a}c_{at-1} + \phi_{2a}c_{at-2} + \varepsilon_{at}, \varepsilon_{at} \sim N(0, \sigma_{\varepsilon_a}^2) \quad (9)$$

First of all, we assume that each variable has a stochastic trend component ($\tau_{it}, i = Y, U \text{ or } A$) and a cyclical component ($c_{it}, i = Y, U \text{ or } A$). We perform the unit root test on each series and find that we can not reject the null hypothesis that all the series have a unit root. Thus a stochastic trend component should be included in the series decomposition. Moreover, it requires that there is no seasonality in the data series, which we verify to be true by performing the seasonality test. We also do not impose the common trend restriction, i.e., all three variables have their own trend and cycle components and these components are allowed to have a certain degree of correlation based on the economic intuition we will discuss later.

Secondly, we specify the dynamics of trend and cycle components. The cyclical component in each series is assumed to follow an AR (2) process. This assumption provides rich dynamics in the data series to enable us to identify all the parameters under the state-space model framework (Morley, Nelson and Zivot, 2003). The trend component are assumed to follow a random walk process with a drift, and as mentioned above, we do not impose a common trend among these three variables.

Thirdly, we assume the shocks to the trend and cycle components to follow a white noise process, but allow for potentially different standard deviations across series. The shocks to the trend components ($\eta_{it}, i = Y, U, \text{ or } A$) have a long-run effect on the trend because the trend follows a random walk process. The shocks to the cyclical component ($\varepsilon_{it}, i = Y, U \text{ or } A$) have a short-run effect on the cycle because the cycle follows a stationary autoregressive process with two lags. The shocks to the trend component are allowed to be correlated across the three variables, so as the shocks to the cyclical components. However, we impose the zero correlation between the shocks to the trend component and the shocks to the cycle component within and between series. That is to say, we assume that the shocks that generate a long-run effect are different from the shocks that generate a short-run effect. This assumption aims to isolate the monetary policy shocks, which often are considered neutral in the long term, from the productivity shocks, which has a persistent effect in the real economy. These shocks are considered to have no linear correlation but still can be dependent with each other in other forms. This assumption is much weaker than assuming

independence among these shocks.

Below is the correlated multivariate unobserved component model setup based on the previous discussion. It is worth to point out that, in the variance-covariance matrix of the shocks to the trend and cycle, $\sigma_{\eta y \eta u}$, $\sigma_{\eta y \eta a}$, and $\sigma_{\eta u \eta a}$ are the pairwise covariance of the shocks to the trend of EME dollar credit, monetary policy rate and dollar index. $\sigma_{\varepsilon y \varepsilon u}$, $\sigma_{\varepsilon y \varepsilon a}$, and $\sigma_{\varepsilon u \varepsilon a}$ are the pairwise covariance of the shocks to the cycle of EME dollar credit, monetary policy rate and dollar index. The estimates of correlation coefficients, instead of covariances, will be reported in Table 1. We are now going to estimate the parameter values using maximum likelihood method (MLE) and decompose the trend and cycle components of each variable by applying the Kalman Filtering procedure in the state-space model framework.³

Measurement Equation:

$$\begin{bmatrix} Y_t \\ U_t \\ A_t \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \tau_{yt} \\ c_{yt} \\ c_{yt-1} \\ \tau_{ut} \\ c_{ut} \\ c_{ut-1} \\ \tau_{at} \\ c_{at} \\ c_{at-1} \end{bmatrix}$$

³See Kim and Nelson (2000) for the details of the estimation procedure.

Transition Equation:

$$\begin{bmatrix} \tau_{yt} \\ c_{yt} \\ c_{yt-1} \\ \tau_{ut} \\ c_{ut} \\ c_{ut-1} \\ \tau_{at} \\ c_{at} \\ c_{at-1} \end{bmatrix} = \begin{bmatrix} \mu_y \\ 0 \\ 0 \\ \mu_u \\ 0 \\ 0 \\ \mu_a \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \phi_{1y} & \phi_{2y} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \phi_{1u} & \phi_{2u} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \phi_{1a} & \phi_{2a} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \tau_{yt-1} \\ c_{yt-1} \\ c_{yt-2} \\ \tau_{ut-1} \\ c_{ut-1} \\ c_{ut-2} \\ \tau_{at-1} \\ c_{at-1} \\ c_{at-2} \end{bmatrix} + \begin{bmatrix} \eta_{yt} \\ \varepsilon_{yt} \\ 0 \\ \eta_{ut} \\ \varepsilon_{ut} \\ 0 \\ \eta_{at} \\ \varepsilon_{at} \\ 0 \end{bmatrix}$$

Variance-Covariance Matrix of the Shocks to Trend and Cycle:

$$Q = \begin{bmatrix} \sigma_{\eta y}^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_{\varepsilon y}^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \sigma_{\eta y \eta u} & 0 & 0 & \sigma_{\eta u}^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_{\varepsilon y \varepsilon u} & 0 & 0 & \sigma_{\varepsilon u}^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \sigma_{\eta y \eta a} & 0 & 0 & \sigma_{\eta u \eta a} & 0 & 0 & \sigma_{\eta a}^2 & 0 & 0 \\ 0 & \sigma_{\varepsilon y \varepsilon a} & 0 & 0 & \sigma_{\varepsilon u \varepsilon a} & 0 & 0 & \sigma_{\varepsilon a}^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4 Results and Interpretation

After the state-space representation of the correlated multivariate UC model, we identify 21 parameters to be estimated using maximum likelihood method. The first subsection will display all the parameter estimates and interpret the correlation of the shocks to the trend and cycle among the three variables. In the second subsection, we follow the Kalman Filtering procedure to obtain the trend and cycle of each series and compare them with the decomposition of the baseline models.

4.1 Dynamic Relationship among EME dollar Credit, US Interest Rate and the Dollar Index

Table 1 provides the maximum likelihood estimates of all the parameters and the corresponding standard errors are in the parentheses. The results suggest that the variation in the overall EME dollar credit is mainly driven by the cyclical shocks. The standard deviation of the shocks in the trend of EME dollar credit is 0.005, much smaller compared to the standard deviation of the shocks in the cycle, 0.02. Additionally, for 1-year treasury constant maturity rate, the standard deviation of the shocks in the trend is 0.278, more than three times as large as the standard deviation of the shocks in the cycle. For the dollar index, the standard deviation of shocks to trend and cycle components are 0.017 and 0.015 respectively, which are very close to each other.

Since our primary goal is to understand the dynamics in EME dollar credit, we may want to take a further look at the pairwise correlation between the shocks in the cycle/trend among these three variables.

The correlation analysis of the shocks to the cycles among these variables suggests that the cyclical variation among EME dollar credit, monetary policy rate and dollar credit are strongly correlated through the short-term shocks. The correlation coefficient between the shocks in the cycle of dollar credit and the shocks in the cycle of monetary rate is around -0.99. The correlation coefficient between the shocks in the cycle of dollar credit and the shocks in the cycle of dollar index is -0.97. And the correlation coefficient between the shocks in the cycle of US interest rate and the

shocks in the cycle of dollar index is 0.97. The standard errors of these estimated parameters confirm that these correlation coefficients are significantly different from zero.

These results are consistent with many stylized facts different researchers have observed. An increase in EME dollar credit are often related to the US monetary policy spillover through the low monetary policy rate. The strong negative correlation between the shocks to the cycle of EME dollar credit and the shocks to the cycle of monetary policy rates provide some evidence to support this conjecture. Furthermore, after the recent global financial crisis, since the US monetary policy rate hit the zero-lower bound and some unconventional monetary policies have been implemented, the US policy rate may not capture the Federal Reserve monetary policy stand as well as before. The dollar index, instead, may be able to capture the effect of unconventional monetary policies, because an expansionary monetary policy will artificially inject the dollar liquidity into the US economy and thus weaken the dollar valuation against other currencies. The weaken dollar are often conjectured to contribute to the rise of dollar credit in EMEs, which is supported by our strongly negative correlation between the short-term shocks to dollar index and EME dollar credit. Finally, the positive correlation between the cyclical movements of the US interest rate and the dollar index supports the idea that an increase in the US monetary policy rate typically strengthens the US dollar exchange rate in the international currency market.

The correlation analysis of the shocks to the trends among these variables suggest that these three variables do not seem to share a common trend.⁴ The correlation coefficient between the shocks in the trend of EME dollar credit and the shocks in the trend of monetary policy rate is 0.62. The correlation coefficient between the shocks in the trend of EME dollar credit and the shocks in the trend of dollar index is 0.61. These estimates may suggest that in the long run, higher interest rate and stronger dollar index indicate the promising economic growth perspective, so the demand for dollar credit in EMEs also increases. The correlation coefficient between the shocks in the trend of interest rate and the shocks in the trend of dollar index is weakly negative at -0.24. The standard errors of these estimated parameters indicate that these correlation coeffi-

⁴In extreme case of cointegration, we will observe perfect correlation in the trend shock of these variables.

Table 1
MAXIMUM LIKELIHOOD ESTIMATES

Description	Parameter	Estimate (Standard Error)
Log likelihood value	llv	335.4485
EME dollar credit		
S.D. of permanent shocks to EME dollar credit	$\sigma_{\eta y}$	0.0051 (0.0025)
S.D. of temporary shocks to EME dollar credit	$\sigma_{\varepsilon y}$	0.0218 (0.0022)
EME dollar credit drift	μ_y	0.0269 (0.0015)
EME dollar credit 1st AR parameter	ϕ_{1y}	1.2393 (0.1112)
EME dollar credit 2nd AR parameter	ϕ_{2y}	-0.3557 (0.1155)
1-year policy rate		
S.D. of permanent shocks to 1-year policy rate	$\sigma_{\eta u}$	0.2776 (0.0266)
S.D. of temporary shocks to 1-year policy rate	$\sigma_{\varepsilon u}$	0.0811 (0.0260)
1-year policy rate drift	μ_u	-0.0501 (0.0395)
1-year policy rate 1st AR parameter	ϕ_{1u}	1.8868 (0.0296)
1-year policy rate 2nd AR parameter	ϕ_{2u}	-0.9279 (0.0286)
US dollar index		
S.D. of permanent shocks to US dollar index	$\sigma_{\eta a}$	0.0167 (0.0035)
S.D. of temporary shocks to US dollar index	$\sigma_{\varepsilon a}$	0.0149 (0.0035)
US dollar index drift	μ_a	-0.0022 (0.0023)
US dollar index 1st AR parameter	ϕ_{1a}	1.2452 (0.1240)
US dollar index 2nd AR parameter	ϕ_{2a}	-0.4896 (0.1038)
Cross-series correlations		
Correlation: Permanent dollar credit/ Permanent interest rate	$\rho_{\eta y \eta u}$	0.6233 (0.5560)
Correlation: Permanent dollar credit/ Permanent dollar index	$\rho_{\eta y \eta a}$	0.6116 (0.6183)
Correlation: Permanent interest rate/ Permanent dollar index	$\rho_{\eta u \eta a}$	-0.2374 (0.2034)
Correlation: Transitory dollar credit/ Transitory interest rate	$\rho_{\varepsilon y \varepsilon u}$	-0.9999 (0.0025)
Correlation: Transitory dollar credit/ Transitory dollar index	$\rho_{\varepsilon y \varepsilon a}$	-0.9663 (0.0542)
Correlation: Transitory interest rate/ Transitory dollar index	$\rho_{\varepsilon u \varepsilon a}$	0.9685 (0.0609)

cients are not significantly different from zero. These results provide the evidence that the shocks that have a long-term impact on these variables are not significantly correlated with each other. Therefore, it is not likely that these variables share the common trend, which supports the conjecture that we need to decompose the trend and cycle components from the series to understand the business cycles of these variables.

4.2 Trend-Cycle Decomposition

In this subsection, we decompose the trend and cycle of EME dollar credit, monetary policy rate and dollar index using the correlated multivariate unobserved component (UC hereafter) model. To check the robustness of our results, we substitute 1-year treasury bill rate with the shadow interest rate calculated based on Wu and Xia (2015) to measure the US policy stance. We also use the decomposition of the same series from various univariate models to serve as the benchmark for comparison.

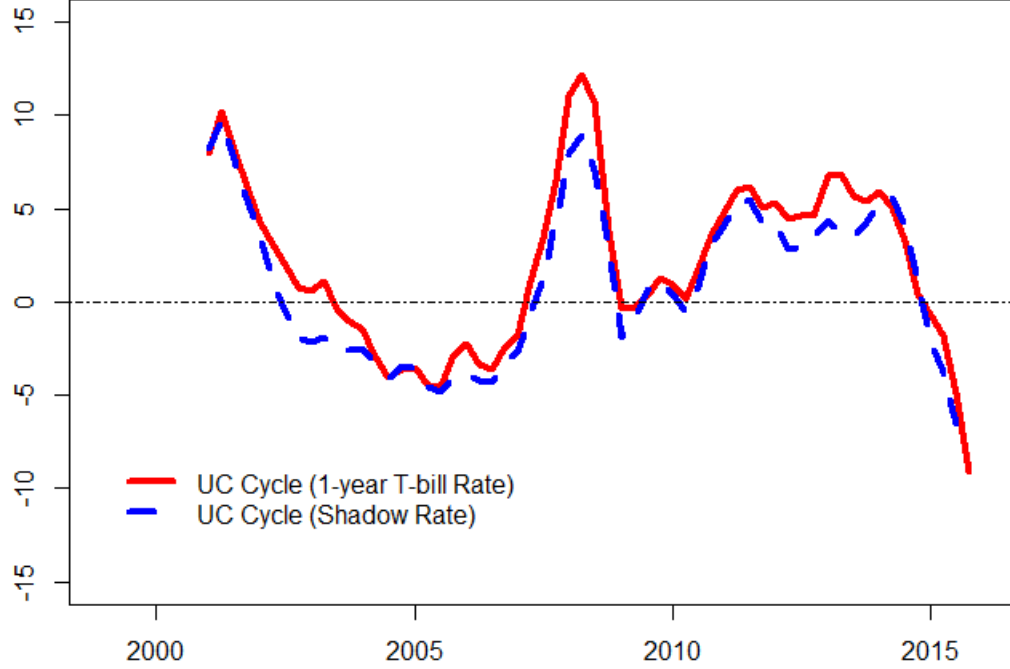
Figure 1: Trend and Cycle Decomposition from Correlated Multivariate UC Model



The stochastic trend in the multivariate UC model captures the long-run increase in the EME dollar credit and also reflects the effect of recent global financial crisis (Figure 1). In the long run, there is an increasing trend in EME dollar credit, due to the global financial integration. There was a downward shift in the trend since the third quarter of 2008, when the sub-prime mortgage crisis turned into a severe global financial crisis, named as Great Recession. The effect of this negative shock on the trend of EME dollar credit persisted for around six years until it phased out in 2015, when the US labor market observed a steady recovery from the recession.

The cyclical component from our multivariate UC model captures the evolution of the dollar credit and its dynamic relationship with US interest rate and the dollar index really well (Figure 1). After the dot-com crisis at the beginning of the century, the US economy steadily recovered with the falling policy rate and then entered the phase featured with housing market boom. In 2007, the sub-prime mortgage crisis triggered housing bubble bust. After the collapse of Bear Steins, emerging market economies provided a sanctuary for the international capital, mainly because the investor community assumed that the EMEs are decoupled from the developed markets. In the third quarter of 2008, Lehman Brother filed bankruptcy and many other financial corporations were also close to be liquidated. These events lead to a global contraction in credit lending, which brought the dollar credit back down to its long-term trend. The Federal Reserve immediately lowered the policy rate and dumped liquidity into the financial market through several runs of Quantitative Easing (QE) programs. The portfolio re-balancing effect quickly directed the investors to the credit market in the EMEs to chase for yield. Shin (2013) denotes the year 2009 as the beginning of the second phase of global liquidity, because of the rapid capital flow into the EMEs in the form of non-financial firm bond issuance. Our cyclical component from the multivariate UC model indicates the expansionary cycle of EME dollar credit during 2009-2014, a phenomenon widely observed and agreed upon among the financial market observers and researchers. Starting from 2015, the persistent recovery of US economy and the slowdown in the EMEs overturned the dollar credit flow and generate the concern about the instability of the EME financial markets and the feedback loops between EMEs and advanced economies.

Figure 2: The Cyclical Component from Correlated Multivariate UC Model



4.2.1 Trend-Cycle Decomposition with Shadow Interest Rate

In our study, we have used 1-year treasury bill rate as a measure of US interest rate. We use this measure because of the zero-lower bound problem that the federal fund rate or 3-month T-bill rate encounter in a big portion of the sample period under our study. The federal fund rate hit the zero-lower bound very soon after the crisis outbreak, followed by the implementation of unconventional monetary policies. Therefore, the academic literature suggest that the federal fund rate has lost the power to accurately measure the Federal Reserve policy stance since then.

One could argue that 1-year T-bill yield does not accurately reflect the stance of monetary policy and may be contaminated with term premia. To take into account the concern, we use an alternative measure called the shadow interest rate as computed in Wu and Xia (2015). The shadow interest rate takes into account the macroeconomic effects of unconventional monetary policy and is not

constrained by the zero-lower bound, based on a nonlinear term structure model. The resulting cycle series ⁵ track closely with the counterpart using 1-year treasury bill rate, confirm the robustness of our model specification (Figure 2).

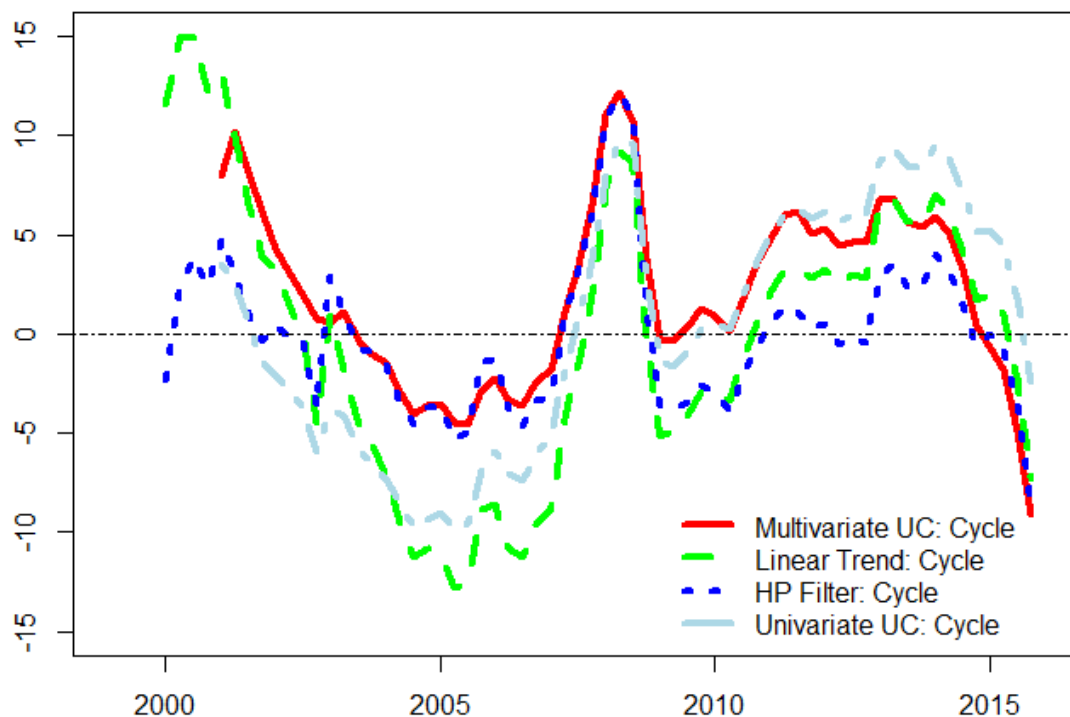
4.2.2 Comparison with Univariate Trend-Cycle Decomposition

We also perform trend-cycle decomposition using univariate models for comparison. The univariate models include linear trend model, HP filter model and univariate UC model. The linear trend model decomposes the EME dollar credit series into a linear-trend component and a cycle component. The HP filter method uses an algorithm to smooth the original data series to estimate the trend component and the difference between them is the cyclical component. The parameter value λ is set at 1600 as suggested by Hodrick and Prescott for the quarterly data. The univariate UC model only uses the series of EME dollar credit to decompose a stochastic trend component and a cyclical component with the same specification as in the multivariate UC model.

The decomposition from the multivariate UC model is more reliable in understanding the trend and cycle behavior throughout the sample periods, based on our findings from the comparison. The linear trend model, assuming a constant slope in the trend component, is not appropriate by ignoring the consequential shock from the Great Recession. The recent crisis is considered as the most severe financial crisis after the Great Depression (1929-1933) and we observe a clearly downward-shifting effect on the EME dollar credit series. Other univariate models, without assuming a linear trend, fail to generate accurate trend and cycle series by ignoring the relationship between EME dollar credit and its price channels. The HP cycle seems to mimic the counterpart from multivariate UC model very well before the crisis but diverge afterwards. The cycle from the univariate UC model, instead, is close to the multivariate counterpart after the crisis. Taking into account all the historical events as mentioned in the cycle interpretation, the outflow of dollar credit slowed down before the crisis, due to the domestic boom, but the magnitude of the negative cycle did not seem to be as large as suggested by the univariate UC model. The HP cycle fails to capture the credit

⁵The parameter estimates from the multivariate UC model with the shadow rate are also similar to the counterparts in the multivariate UC model with 1-year treasury bill rate.

Figure 3: Cycles in Multivariate UC Decomposition / Univariate Decomposition

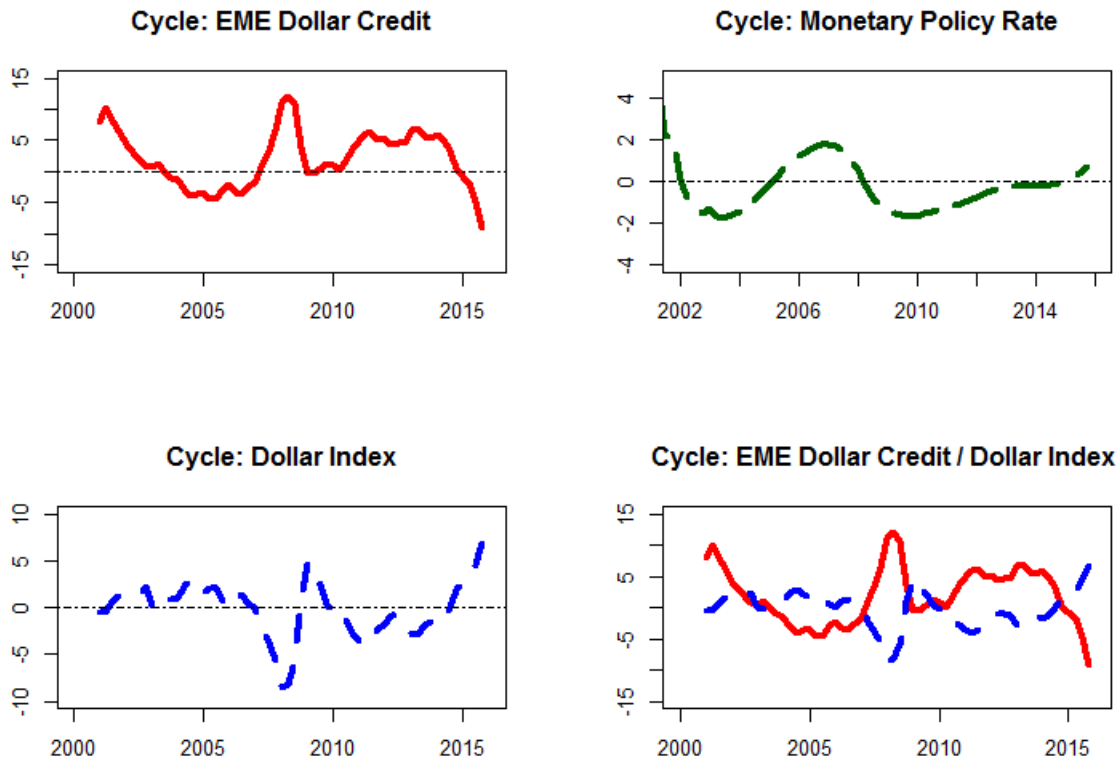


boom since 2009, while it performed reasonably well before the crisis. Overall, we believe that our correlated UC model out-performs the univariate models in the whole sample periods in reflecting the trend and cycle variation in EME dollar credit.

4.2.3 Co-movement among the Cyclical Components

The cyclical components of the three variables in our multivariate unobserved component model uncover the fact that the cyclical variation of dollar index is strongly negatively associated with the cyclical variation of EME dollar credit, an indicator containing more useful information than the US monetary policy rate variable. Figure 4 displays the cycles of EME dollar credit, 1-year treasury bill rate and dollar index. Before 2008 financial crisis, when dollar was relatively strong,

Figure 4: Cycles in Multivariate UC Decomposition / Univariate Decomposition



the overall EME dollar credit was increasing but was still below its trend, because the cyclical dollar appreciation creates the adverse force to this capital inflow in EMEs. The cyclical component suggests that the strong dollar could potentially contribute to slowing down the credit flow into the EMEs. When the financial crisis hit the US economy, the dollar was severely weakened and the investors with dollar assets were looking for higher yields in alternative asset classes. During this recessionary phase in the US, the EME dollar credit witnessed an expansionary period during the 2009-2014 sample period. Since 2015, the recovery of US economy strengthened the US dollar in the currency market and the probability of an interest rate hike by the Federal Reserve has gone up. The EME dollar credit seemed to have entered a contractionary phase. Note that the overall EME dollar credit could still increase due to the underlying increasing trend.

The dollar credit cycle is weakly negatively associated with the cyclical component of monetary

policy rate in US. This result is consistent with many other studies which found the negative correlation between these two variables, but were unable to attain the high levels of statistical significance. This phenomenon can be explained by the following. As a monetary policy instrument, US monetary policy rate is less volatile compared to the market-based variables like EME dollar credit or dollar index. Moreover, the policy rate cannot capture the effect of unconventional monetary policy, for instance, the large bond purchase activities. Furthermore, the dollar index is more relevant for the market participants to estimate the overall return when the US policy rate hits the zero-lower bound. The risk of international carry trade, a typical cyclical investment behavior, mainly comes from the exchange rate risk under this environment. Therefore, it is not surprising that the dollar index can provide useful information in understanding the EME dollar credit condition.

5 Conclusions

In this paper, we use a correlated multivariate unobserved component model to decompose the movements in EME dollar credit, monetary policy rate and dollar index into permanent and transitory components. In doing so, we also estimate the correlation between transitory and permanent shocks of each series. The estimated trend and cyclical component from our model captures the dynamic features of EME dollar credit series and performs better than univariate benchmarks in capturing the boom and the boost during the last few years. The correlations among the cyclical components support the idea that the rise of dollar credit in EMEs is associated with low US interest rate and dollar depreciation. The strong cyclical correlations among dollar credit in EMEs, US interest rate and the dollar index suggest that the policy makers may need to take into account the US monetary policy spillover effect on domestic credit conditions of the EMEs, by observing the stance of the US monetary policy and the behavior of the US dollar in the foreign exchange market.

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